SCHOOL OF MATHEMATICAL AND COMPUTER SCIENCES

Actuarial Mathematics and Statistics

F78PB

PROBABILITY AND STATISTICS B

(MAY 2024)

SOLUTIONS

1. a) The mgf of X is given by:

$$\begin{aligned}
M_X(\theta) &= E\left(e^{\theta x}\right) \\
&= \int_0^\infty e^{\theta x} f(x) dx \\
&= \int_0^\infty e^{\theta x} \left[e^{-2x} + \frac{1}{2} e^{-x}\right] dx \\
&= \int_0^\infty e^{-2x+\theta x} dx + \int_0^\infty \frac{1}{2} e^{-x+\theta x} dx \\
&= \int_0^\infty e^{-x(2-\theta)} dx + \frac{1}{2} \int_0^\infty e^{-x(1-\theta)} dx \\
&= \frac{1}{\theta - 2} \left[e^{-x(2-\theta)}\right]_0^\infty + \frac{1}{2(\theta-1)} \left[e^{-x(1-\theta)}\right]_0^\infty \\
&= \frac{1}{\theta - 2} \left[0 - 1\right] + \frac{1}{2(\theta-1)} \left[0 - 1\right] \\
&= \frac{1}{2-\theta} + \frac{1}{2(1-\theta)} \text{ for } \theta < 1
\end{aligned}$$

[4 marks]

b) The mean and variance of X are given by:

$$\begin{aligned}
E(X) &= M_X'(\theta = 0) \\
M_X'(\theta) &= \frac{1}{(2-\theta)^2} + \frac{1}{2(1-\theta)^2} \\
M_X'(\theta = 0) &= \frac{1}{(2-0)^2} + \frac{1}{2(1-0)^2} \\
&= \frac{1}{4} + \frac{1}{2} \\
&= \frac{3}{4}
\end{aligned}$$

[3 marks]

$$
\begin{aligned}
\mathrm{Var}(X) &= E\left(X^2\right) - [E(X)]^2 \\
E\left(X^2\right) &= M_X''(\theta = 0) \\
M_X''(\theta) &= \frac{2}{(2-\theta)^3} + \frac{1}{(1-\theta)^3} \\
M_X''(\theta = 0) &= \frac{2}{(2-0)^3} + \frac{1}{(1-\theta)^3} \\
&= \frac{2}{8} + 1 \\
&= \frac{5}{4} \\
\mathrm{Var}(X) &= \frac{5}{4} - \left(\frac{3}{4}\right)^2 \\
&= \frac{11}{16}
\end{aligned}
$$

[4 marks]

c) The mgf of Y.

$$
\begin{aligned}
Y &= \frac{1}{2}X \\
M_Y(\theta) &= E\left[e^{\theta Y}\right] \\
&= E\left[e^{\theta \frac{X}{2}}\right] \\
&= E\left[e^{\frac{\theta}{2} X}\right] \\
&= M_X\left(\frac{\theta}{2}\right)
\end{aligned}
$$

[2 marks]

$$
\begin{aligned}
M_Y(\theta) &= M_X\left(\frac{\theta}{2}\right) \\
M_Y(\theta) &= \frac{1}{\left(2 - \frac{\theta}{2}\right)} + \frac{1}{2\left(1 - \frac{\theta}{2}\right)} \\
&= \frac{2}{(4-\theta)} + \frac{1}{(2-\theta)}
\end{aligned}
$$

[2 marks]

2. a) The bias for the estimators are:

$$\begin{aligned} Bias(\hat{\lambda}_1) &= E(\hat{\lambda}_1) - \lambda \\ E(\hat{\lambda}_1) &= E\left[\frac{1}{8}(X_1 + X_2 + X_3 + X_4) + \frac{1}{2}(X_5)\right] \\ &= \frac{1}{8}E(X_1 + X_2 + X_3 + X_4) + \frac{1}{2}E(X_5) \\ &= \frac{4\lambda}{8} + \frac{\lambda}{2} \\ &= \frac{1}{2}\lambda + \frac{1}{2}\lambda \\ &= \lambda \\ \therefore Bias(\hat{\lambda}_1) &= \lambda - \lambda = 0 \end{aligned}$$

[2 marks]

$$\begin{aligned} Bias(\hat{\lambda}_2) &= E(\hat{\lambda}_2) - \lambda \\ E(\hat{\lambda}_2) &= E\left[\frac{X_1 + 2X_2 + 3X_3 + 4X_4 + 5X_5}{5}\right] \\ &= \frac{\lambda + 2\lambda + 3\lambda + 4\lambda + 5\lambda}{5} \\ &= \frac{15\lambda}{5} \\ &= 3\lambda \\ \therefore Bias(\hat{\lambda}_2) &= 3\lambda - \lambda = 2\lambda \end{aligned}$$

[2 marks]

Comparing the two estimators, $\hat{\lambda}_1$ is considered to be better since it is an unbiased estimator, whilst $\hat{\lambda}_2$ is a biased estimator of λ.

[1 mark]

b) The MSE of $\hat{\lambda}_2$

$$MSE(\hat{\lambda}_2) = Var(\hat{\lambda}_2) + bias^2(\hat{\lambda}_2)$$

$$\begin{aligned} Var(\hat{\lambda}_2) &= Var\left[\frac{1}{5}(X_1 + 2X_2 + 3X_3 + 4X_4 + 5X_5)\right] \\ &= \frac{1}{25}\left[Var(X_1) + 4Var(X_2) + 9Var(X_3) + 16Var(X_4) + 25Var(X_5)\right] \\ &= \frac{1}{25}\left[\lambda^2 + 4\lambda^2 + 9\lambda^2 + 16\lambda^2 + 25\lambda^2\right] \\ &= \frac{11\lambda^2}{5} \end{aligned}$$

[2 marks]

$$\therefore MSE(\hat{\lambda}_2) = \frac{11\lambda^2}{5} + (2\lambda)^2 = \frac{31\lambda^2}{5}$$

[1 mark]

c) $\hat{\lambda}_1$ is a more efficient estimator than $\hat{\lambda}_2$ because the MSE of $\hat{\lambda}_1$ smaller compared to the MSE of $\hat{\lambda}_2$. That is,

$$\frac{MSE(\hat{\lambda}_1)}{MSE(\hat{\lambda}_2)} = \frac{5\lambda^2}{16}.\frac{5}{31\lambda^2} = \frac{25}{496} < 1$$

[2 mark]

3. a)

$$\begin{aligned} E(Y) &= \int_{-\infty}^{\infty} yf(y)dy \\ &= \int_0^1 y(1-\theta) + 2\theta y^2 dy \\ &= \left[\frac{y^2(1-\theta)}{2} + \frac{2\theta y^3}{3}\right]_0^1 \\ &= \frac{1}{2}(1-\theta) + \frac{2\theta}{3} \\ &= \frac{1}{2} - \frac{\theta}{2} + \frac{2\theta}{3} \\ &= \frac{\theta+3}{6} \end{aligned}$$

[3 marks]

The MME of θ is:

$$\begin{aligned} \bar{y} &= \frac{\theta+3}{6} \\ 6\bar{y} - 3 &= \theta \\ \hat{\theta} &= 6\bar{y} - 3 \end{aligned}$$

[2 marks]

b)

$$\begin{aligned} L(\lambda; x_i) &= \prod_{i=1}^{3} f(x_i) \\ &= \lambda e^{-8\lambda} \cdot \lambda e^{-6\lambda} \cdot \lambda e^{-3\lambda} \\ &= \lambda^3 e^{-\lambda(8+6+3)} \\ &= \lambda^3 e^{-17\lambda} \\ logL(\lambda; x_i) &= 3\log(\lambda) - 17\lambda \\ \frac{\partial logL(\lambda; x_i)}{\partial \lambda} &= \frac{3}{\lambda} - 17 = 0 \\ \hat{\lambda} &= \frac{3}{17} \end{aligned}$$

[6 marks]

4. a) $\frac{(\bar{x}_A - \bar{x}_B) - (\mu_A - \mu_B)}{\sqrt{S_p^2\left(\frac{1}{n_A} + \frac{1}{n_B}\right)}}$ is a pivotal quantity for $(\mu_A - \mu_B)$ because:

   i. It is a function which involves the parameters μ_A and μ_B, and observations
   ii. It has a known distribution which does not depend on the parameters μ_A and μ_B. That is, t distribution with $(n_A + n_B - 2)$ degrees of freedom.
   iii. It is a monotonic function of the parameters

[3 marks]

b) Since the 95% confidence interval (-1551.37, -1425.44) does not contain zero, we can conclude that the mean volumes produced differ significantly between the two energy drinks produced.

[2 marks]

c)

$$\begin{aligned} F_{\alpha/2,(n_A-1),(n_B-1)} &= F_{0.025,(11),(11)} = 3.47 \\ F_{\alpha/2,(n_B-1),(n_A-1)} &= F_{0.025,(11),(11)} = 3.47 \end{aligned}$$

[2 marks]

Therefore, a 95% CI for $\frac{\sigma_A^2}{\sigma_B^2}$, the ratio of the variances for the volumes produced is given by:

$$\left(\frac{S_A^2}{S_B^2} \cdot \frac{1}{F_{\alpha/2,(n_A-1),(n_B-1)}}, \frac{S_A^2}{S_B^2} \cdot F_{\alpha/2,(n_B-1),(n_A-1)}\right)$$

$$\left(\left(\frac{5000}{6100}\right)\left(\frac{1}{3.47}\right), \left(\frac{5000}{6100}\right)(3.47)\right)$$

$$(0.236, 2.844)$$

[2 marks]

Conclusion:
The interval (0.236,2.844) contains one. Therefore, the company may conclude that the variance in the volumes of drinks produced do not differ significantly at 95% confidence level. This is important in order to choose appropriate method for constructing the confidence interval for mean volumes difference for the two energy drinks produced.

[1 mark]

5. **Hypothesis:**

$$H_0 : \mu_X = \mu_Y$$
$$H_1 : \mu_X > \mu_Y$$

Critical region: Reject H_0 if $t > t_{\alpha,(n_X+n_Y-2)}$

$$t_{\alpha,(n_X+n_Y-2)} = t_{0.05(27)} = 1.703$$

[1 mark]

Test statistic:

$$S_p^2 = \frac{(n_X-1)S_X^2 + (n_Y-1)S_Y^2}{n_X+n_Y-2} = 1061.42$$
$$t = \frac{(\bar{x}-\bar{y}) - (\mu_X-\mu_Y)}{S_p\sqrt{\frac{1}{n_X}+\frac{1}{n_Y}}} = 5.573$$

[3 marks]

Decision and conclusion:
Since the test statistic $t = 5.573 > 1.703$, we reject H_0 and conclude that $\mu_X > \mu_Y$ at 5% significance level.

[2 marks]

6. a)

$$\bar{x} = \frac{1}{n}\sum_{i=1}^{n} x_i = 25.967$$

$$\bar{y} = \frac{1}{n}\sum_{i=1}^{n} y_i = 24.767$$

$$S_{XX} = \sum_{i=1}^{n} x_i^2 - \frac{1}{n}\left(\sum_{i=1}^{n} x_i\right)^2 = 43.087$$

$$S_{YY} = \sum_{i=1}^{n} y_i^2 - \frac{1}{n}\left(\sum_{i=1}^{n} y_i\right)^2 = 47.147$$

$$S_{XY} = \sum_{i=1}^{n} x_i y_i - \frac{1}{n}\left(\sum_{i=1}^{n} x_i\right)\left(\sum_{i=1}^{n} y_i\right) = -29.493$$

$$\hat{\beta}_1 = \frac{S_{XY}}{S_{XX}} = \frac{-29.493}{43.087} = -0.684$$

$$\hat{\beta}_0 = \bar{y} - \hat{\beta}_1 \bar{x} = 24.767 - (-0.684)(25.967) = 42.528$$

[2 marks]

Therefore, the least squares regression line is given by:

$$\hat{y}_i = 42.528 - 0.684 x_i, \text{ for } i = 1, 2, \ldots, 12$$

From the estimated model, the shear resistance (y) of the spacemen is expected to decrease by 0.684 for a unit change in the normal stress (x).

[1 mark]

b) **Hypothesis:**

$$H_0 : \beta_1 = 0$$
$$H_1 : \beta_1 \neq 0$$

Critical region: reject H_0 if $|t| > t_{0.025(10)} = 2.228$

[1 mark]

Test statistic:

$$S^2 = \frac{S_{YY} - \hat{\beta}_1 S_{XY}}{n-2} = 2.697$$

$$t = \frac{\hat{\beta}_1 - \beta_1}{\sqrt{\frac{S^2}{S_{XX}}}} = \frac{-0.684 - 0}{\sqrt{\frac{2.697}{43.087}}} = -2.734$$

[1 mark]

Decision and conclusion:
Since the test statistic $|t| = 2.734 > 2.228$, we reject H_0 and conclude that $\hat{\beta}_1$ is not equal to zero at 5% significance level.

[1 mark]

c) The three model assumptions that are being check with the residual plot are:

i. The residuals are normally distributed with mean zero
ii. The variance of the residuals is contant (homoscedasticity)
iii. The residuals are independent of each other

[2 marks]

[Grand Total: 60 marks]